Mathematics · Practice Test 3

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: 0.6, 62%, ⅝, √0.4. A. 0.6, 62%, ⅝, √0.4 B. √0.4, ⅝, 62%, 0.6 C. 62%, 0.6, ⅝, √0.4 D. ⅝, 0.6, 62%, √0.4

2. Between which two whole numbers does √115 lie? A. 9 and 10 B. 10 and 11 C. 11 and 12 D. 57 and 58

3. *(Griddable)* Write 5,200,000 in scientific notation. ☐

4. Which number is **irrational**? A. √45 B. √49 C. ¾ D. 0.5

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−1, −2) and (3, 6)? A. 2 B. 4 C. 8 D. ½

6. A proportional relationship passes through the origin and (4, 10). What is the unit rate? A. 2 B. 2.5 C. 4 D. 6

7. *(Equation editor)* A pool fills at 25 liters per minute, starting empty. Write an equation for the amount y after x minutes. ☐

8. Which relation is a **function**? A. {(3, 1), (3, 2), (4, 5)} B. {(−1, 2), (0, 2), (1, 3)} C. {(2, 4), (2, 5), (3, 6)} D. {(7, 1), (7, 2), (8, 3)}

9. Which equation represents a **non-proportional** relationship? A. y = 5x B. y = ⅓x C. y = 2x − 5 D. y = 8x

10. *(Griddable)* For a direct variation, y = 48 when x = 6. What is y when x = 5? ☐

11. Which table shows a **proportional** relationship? A. (1, 3)(2, 6)(3, 9) B. (1, 2)(2, 5)(3, 8) C. (1, 4)(2, 6)(3, 8) D. (2, 9)(3, 11)(4, 13)

12. *(Griddable)* Solve for x: 9x − 7 = 4x + 18. ☐

13. "Seven less than four times a number equals the number plus eight." Which equation models this? A. 4x − 7 = x + 8 B. 7 − 4x = x + 8 C. 4x + 7 = x − 8 D. 7x − 4 = x + 8

14. Two lines intersect at (4, −1). What is the solution of the system? A. x = −1, y = 4 B. x = 4, y = −1 C. x = 3 D. No solution

15. The slope between any two points on a line is constant because the right triangles formed are — A. congruent B. **similar** C. equilateral D. right isosceles

16. Which situation is **proportional**? A. $30 plus $10 per hour B. 4 pounds of rice per $6, at a steady rate C. 50 L draining 4 L/min D. $12 fee plus $3 each

17. Which situation can be modeled by $6x + 2 = 2x + 22$? A. Two savers start with $2 and $22, saving $6 and $2 per week; when equal? B. 6 times 2 C. A triangle's area D. 2 plus 22

18. A bike rental costs $5 plus $2 per hour. Which equation gives the cost y for x hours? A. $y = 5x + 2$ B. $y = 2x + 5$ C. $y = 7x$ D. $y = 2x - 5$

19. *(Select TWO)* Which statements about $y = 4x - 7$ are true? A. The slope is 4 B. The slope is −7 C. The y-intercept is −7 D. It passes through the origin

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 3 and height 10. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 141.3 C. 282.6 D. 565.2

21. A cone has radius 3 and height 7. What is its volume? ($\pi \approx 3.14$) A. 21.98 B. 65.94 C. 131.88 D. 197.82

22. A cylinder has radius 4 and height 5. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 150.72 C. 226.08 D. 351.68

23. *(Griddable)* A right triangle has legs of 8 and 15. Find the hypotenuse. ☐

24. What is the distance between (1, 2) and (5, 5)? A. 5 B. 7 C. 12 D. 25

25. The point (3, −5) is dilated by a scale factor of 2 about the origin. What is the image? A. (5, −3) B. (6, −10) C. (1.5, −2.5) D. (6, −5)

26. The point (−2, −7) is reflected over the y-axis. What is the image? A. (2, −7) B. (−2, 7) C. (2, 7) D. (7, −2)

27. The point (−3, 4) is rotated 180° about the origin. What is the image? A. (3, −4) B. (3, 4) C. (−4, 3) D. (4, −3)

28. A triangle has area 10. It is dilated by a scale factor of 3. What is the area of the image? A. 13 B. 30 C. 90 D. 100

29. An exterior angle of a triangle equals the sum of its two remote interior angles, 40° and 75°. What is the exterior angle? A. 65° B. 105° C. 115° D. 125°

30. Two similar figures have corresponding sides in a ratio of 5 : 8. A side of the smaller is 15. What is the matching side of the larger? A. 18 B. 21 C. 24 D. 27

31. *(Griddable)* The base of a cylinder is a circle with radius 5. What is the base area B? ($\pi \approx 3.14$) ☐

32. Which transformation **changes** the size of a figure? A. Translation B. Reflection C. Rotation D. Dilation by factor 4

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 5, 7, 9, 11. ☐

34. A trend line is modeled by $y = 2x + 1$. Predict y when $x = 12$. A. 13 B. 23 C. 25 D. 27

35. On a scatterplot, points rise from lower-left to upper-right. What association is shown? A. Positive linear B. Negative linear C. No association D. Non-linear

36. Which pattern suggests the data are **linear**? A. A U-shaped curve B. A circular cluster C. A straight rising path D. Random scatter

37. *(Griddable)* Find the simple interest on $1,500 at 4% for 5 years. ☐

38. For the same amount and rate, which loan length costs the **most** total interest? A. 1 year B. 2 years C. 4 years D. 6 years

39. How do small, regular investments behave over many years? A. Shrink B. Stay equal C. Grow, especially with compound interest D. Disappear

40. College costs $20,000. A family contributes $8,000 and saves the rest over 24 months. How much per month? A. $400 B. $500 C. $600 D. $750

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.

Mathematics · Practice Test 4

40 questions · Calculator allowed · Use the Grade 8 Mathematics Reference Sheet

Read each question carefully. Watch for directions like *Select TWO* and *least to greatest*.

Reporting Category 1 · Numerical Representations and Relationships

1. Order from **least to greatest**: ⅞, 0.88, 89%, √0.8. A. ⅞, 0.88, 89%, √0.8 B. √0.8, 89%, 0.88, ⅞ C. 0.88, ⅞, 89%, √0.8 D. 89%, ⅞, 0.88, √0.8

2. Between which two whole numbers does √150 lie? A. 11 and 12 B. 12 and 13 C. 13 and 14 D. 74 and 76

3. *(Griddable)* Write 0.000072 in scientific notation. ☐

4. Which number is **irrational**? A. √24 B. √25 C. ⅖ D. 0.1

Reporting Category 2 · Computations and Algebraic Relationships

5. What is the slope of the line through (−3, 4) and (1, −4)? A. 2 B. −2 C. ½ D. −½

6. A proportional relationship passes through the origin and (6, 9). What is the unit rate? A. 1.5 B. 2 C. 3 D. 6

7. *(Equation editor)* A savings account starts with $40 and grows $15 each week. Write an equation for the balance y after x weeks. ☐

8. Which relation is a **function**? A. {(1, 2), (1, 5), (3, 4)} B. {(2, 3), (4, 3), (6, 5)} C. {(0, 1), (0, 4), (2, 6)} D. {(3, 1), (3, 9), (5, 2)}

9. Which equation represents a **non-proportional** relationship? A. y = 5x + 2 B. y = 6x C. y = ¼x D. y = 9x

10. *(Griddable)* For a direct variation, y = 54 when x = 6. What is y when x = 8? ☐

11. Which table shows a **proportional** relationship? A. (1, 5)(2, 10)(3, 15) B. (1, 3)(2, 7)(3, 11) C. (1, 2)(2, 5)(3, 9) D. (2, 9)(3, 13)(4, 17)

12. *(Griddable)* Solve for x: 8x + 3 = 5x + 27. ☐

13. "Nine more than twice a number equals five times the number minus six." Which equation models this? A. 2x + 9 = 5x − 6 B. 9 − 2x = 5x − 6 C. 2x − 9 = 5x + 6 D. 9x + 2 = 5x − 6

14. Two lines intersect at (−2, −5). What is the solution of the system? A. x = −5, y = −2 B. x = −2, y = −5 C. x = −7 D. No solution

15. The slope of a line is the same everywhere because the right triangles drawn to its points are — A. congruent B. **similar** C. obtuse D. equilateral

16. Which situation is **proportional**? A. $45 plus $20 per hour B. 3 apples for $1.50 at a steady rate C. 60 L losing 6 L/min D. $9 fee plus $4 each

17. Which situation can be modeled by $7x + 5 = 3x + 25$? A. Two savers start with $5 and $25, saving $7 and $3 per week; when equal? B. 7 times 5 C. A circle's area D. 5 plus 25

18. A phone plan costs $10 plus $0.05 per text. Which equation gives the cost y for x texts? A. $y = 10x + 0.05$ B. $y = 0.05x + 10$ C. $y = 10.05x$ D. $y = 0.05x - 10$

19. *(Select TWO)* Which statements about $y = -2x + 8$ are true? A. The slope is -2 B. The slope is 8 C. The y-intercept is 8 D. The y-intercept is -2

Reporting Category 3 · Geometry and Measurement

20. A cylinder has radius 6 and height 5. What is its volume? ($\pi \approx 3.14$) A. 94.2 B. 282.6 C. 565.2 D. 1130.4

21. A sphere has radius 6. What is its volume? ($\pi \approx 3.14$) A. 150.72 B. 452.16 C. 904.32 D. 1808.64

22. A cylinder has radius 5 and height 4. What is its **total** surface area? ($\pi \approx 3.14$) A. 125.6 B. 219.8 C. 282.6 D. 314

23. *(Griddable)* A right triangle has legs of 20 and 21. Find the hypotenuse. ☐

24. What is the distance between $(-2, -1)$ and $(4, 7)$? A. 6 B. 8 C. 10 D. 14

25. The point $(-4, 6)$ is dilated by a scale factor of ½ about the origin. What is the image? A. $(-2, 3)$ B. $(-8, 12)$ C. $(-2, 6)$ D. $(-4, 3)$

26. The point $(3, -8)$ is reflected over the x-axis. What is the image? A. $(-3, -8)$ B. $(3, 8)$ C. $(-3, 8)$ D. $(8, 3)$

27. The point $(5, 2)$ is rotated 90° counterclockwise about the origin. What is the image? A. $(-2, 5)$ B. $(2, -5)$ C. $(-5, -2)$ D. $(2, 5)$

28. The angles of a triangle measure $2x$, $3x$, and $4x$. What is the **largest** angle? A. 20° B. 40° C. 60° D. 80°

29. A rectangle measures 6 by 4. It is dilated by a scale factor of ½. What is the area of the image? A. 3 B. 6 C. 12 D. 24

30. Two similar triangles have corresponding sides in a ratio of 4 : 7. A side of the smaller is 12. What is the matching side of the larger? A. 18 B. 20 C. 21 D. 24

31. In a right triangle, the square on the hypotenuse equals — A. the sum of the squares on the two legs B. the difference of the legs C. twice the longer leg D. the product of the legs

32. Which transformation **preserves** congruence? A. Dilation by factor 2 B. Dilation by factor 3 C. Rotation of 90° D. Dilation by factor ½

Reporting Category 4 · Data Analysis and Personal Financial Literacy

33. *(Griddable)* Find the mean absolute deviation of 6, 10, 10, 14. ☐

34. A trend line is modeled by $y = 0.5x + 3$. Predict y when $x = 20$. A. 10 B. 13 C. 16 D. 23

35. A scatterplot shows points with no clear direction or pattern. What association is shown? A. Positive linear B. Negative linear C. No association D. Strong linear

36. Which pattern suggests a **non-linear** association? A. A straight rising line B. A straight falling line C. A curved path D. Points on one line

37. *(Griddable)* Find the simple interest on $2,000 at 3% for 6 years. ☐

38. Two accounts hold the same amount at the same rate for the same time. Which earns **more**? A. The simple-interest account B. The compound-interest account C. They earn the same D. Neither earns interest

39. What helps savings grow the most over time? A. Spending it quickly B. Saving once, then stopping C. Regular saving plus compound interest D. Keeping cash at home

40. College costs $24,000. A family contributes $9,000 and saves the rest over 30 months. How much per month? A. $450 B. $500 C. $600 D. $750

Answers and full explanations for this test are in Part 3 · Answer Keys & Explanations.